Alexander Shaham

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EDUCATION

University of California, Santa Cruz — B.S. Computer Science: Computer Game Design — Sep 2017 – Jun 2022

GPA: 3.35 overall; 3.59 over the final two years (2020–2022) · Dean's Honors

Computer engineering and robotics: Robot Automation, Computer Systems and Assembly Language, Applied Discrete Mathematics, Introduction to Networking

Computer science: Data Structures and Algorithms, Advanced Programming, Web Applications, Game Technologies, Game Systems

Math and physics: Calculus I–II, Vector Calculus, Mathematical Methods for Engineers I–II, Statistics, Physics I and III

EXPERIENCE

Chuken Studios — Independent Game Developer — Feb 2026 – Present

Runefall: classless arena-and-dungeon RPG for PC · Godot 4, C#, .NET 10, Steamworks · playable online

- Built an engine-free, server-authoritative 60 Hz simulation with deterministic replays. Snapshots are filtered by each team's line of sight, so hidden entities never leave the server, and matches run online over Steam Datagram Relay.
- Built the combat runtime (ability state machine, projectiles, interrupts, crowd control with diminishing returns) and bot AI that sees only its own team's vision.
- Backed it with 5,000+ xUnit tests, including golden replays verified by state hash and zero-allocation checks.
- Prototyped in Unreal Engine 5 (C++), then migrated to Godot 4 and C# (.NET 10). Manage a ten-milestone roadmap and 450+ tracked tickets.

HCL BigFix — Frontend Software Engineer (SaaS) — Aug 2023 – Feb 2025

- Built and maintained a React + MUI component library adopted by 100+ engineers in 7+ apps.
- Shipped 100+ versioned, documented components at 95%+ test coverage, plus a Storybook-style workbench where engineering, QA, and design reviewed components together.
- Mentored 4 engineers across teams on onboarding, design-system usage, and architectural standards.

Full-stack Software Engineering Intern (WebUI) — May 2023 – Aug 2023

- Integrated Tenable SC and IO into the security-data ingestion pipeline (Node.js, SQL, REST).

Freelance — Web Developer — 2022 – 2023

- Built an accessible, multilingual landing page for a healthcare clinic in React and JavaScript.

King Nutronics Corporation — QA Engineering Intern — Jun 2018 – Sep 2018

- Automated calibration test beds through CAD and microcontroller integration, cutting QA testing time by ~80%.
- Tested calibration hardware used by the U.S. Navy for MIL-STD compliance; found and corrected about 20 critical errors in documentation and logging.

PROJECTS

Guild Golem | TypeScript, React Router, Hono, Cloudflare Workers, D1, R2 — guildgolem.com

- Guild management site for World of Warcraft: Forever, with a raid calendar and Discord sign-ups, raid planner, roster and attendance, loot by council, EPGP, or DKP, and a 13-command Discord bot.
- Rebuilt my earlier Logs and Sims apps inside it, with combat logs parsed in a Web Worker and a sim for 23 specs on game-client data. It has 1,000+ tests and a strict CSP, and bundle budgets are checked before every deploy.

Drafts | TypeScript, React, Hono, Durable Objects, D1, KV, Workers AI — drafts.ajshaham.workers.dev

- Real-time League and Dota 2 draft rooms that suggest picks and bans from matchups and synergies, with an AI draft coach.

TECHNICAL SKILLS

Languages	C#, C++, TypeScript, JavaScript, Python, SQL, Java
Systems	Deterministic simulation, real-time networking, state machines, bot AI
Engines	Godot 4 (.NET), Unreal Engine 5 (C++), Unity
Web & Tools	React, Node.js, PostgreSQL, Cloudflare Workers, Git, GitHub Actions
Spoken	English (native), Spanish (proficient)